254A, Supplement 5: The linear sieve and Chen's theorem (optional)

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We continue the discussion of sieve theory from Notes 4, but now specialise to the case of the *linear sieve* in which the sieve dimension κ is equal to 1, which is one of the best understood sieving situations, and one of the rare cases in which the precise limits of the sieve method are known. A bit more specifically, let $z, D \geq 1$ be quantities with $z = D^{1/s}$ for some fixed $s > 1$, and let g be a multiplicative function with

$$g(p) = \frac{1}{p} + O(\frac{1}{p^2}) \tag{1}$$

and

$$0 \leq g(p) \leq 1 - c \tag{2}$$

for all primes p and some fixed $c > 0$ (we allow all constants below to depend on c). Let $P(z) := \prod_{p<z} p$, and for each prime $p < z$, let E_p be a set of integers, with $E_d := \bigcap_{p \mid d} E_p$ for $d \mid P(z)$. We consider finitely supported sequences $(a_n)_{n \in \mathbf{Z}}$ of non-negative reals for which we have bounds of the form

$$\sum_{n \in E_d} a_n = g(d)X + r_d. \tag{3}$$

for all square-free $d \leq D$ and some $X > 0$, and some remainder terms r_d. One is then interested in upper and lower bounds on the quantity

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n.$$

The fundamental lemma of sieve theory (Corollary 19 of Notes 4) gives us the bound

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n = (1 + O(e^{-s}))XV(z) + O(\sum_{d \leq D : \mu^2(d) = 1} |r_d|) \tag{4}$$

where $V(z)$ is the quantity

$$V(z) := \prod_{p<z} (1 - g(p)). \tag{5}$$

This bound is strong when s is large, but is not as useful for smaller values of s. We now give a sharp bound in this regime. We introduce the functions $F, f : (0, +\infty) \to \mathbf{R}^+$ by

$$F(s) := 2e^\gamma (\frac{1_{s>1}}{s} + \sum_{j \geq 3, \text{ odd}} \frac{1}{j!} \int_{[1,+\infty)^{j-1}} 1_{t_1 + \ldots + t_{j-1} \leq s-1} \frac{dt_1 \ldots dt_{j-1}}{t_1 \ldots t_j}) \tag{6}$$

and

$$f(s) := 2e^\gamma \sum_{j \geq 2, \text{ even}} \frac{1}{j!} \int_{[1,+\infty)^{j-1}} 1_{t_1 + \ldots + t_{j-1} \leq s-1} \frac{dt_1 \ldots dt_{j-1}}{t_1 \ldots t_j} \tag{7}$$

where we adopt the convention $t_j := s - t_1 - \ldots - t_{j-1}$. Note that for each s one has only finitely many non-zero summands in (6), (7). These functions are closely related to the Buchstab function ω from Exercise 28 of Supplement 4; indeed from comparing the definitions one has

$$F(s) + f(s) = 2e^\gamma \omega(s)$$

for all $s > 0$.

Exercise 1 (Alternate definition of F, f) Show that $F(s)$ is continuously differentiable except at $s = 1$, and $f(s)$ is continuously differentiable except at $s = 2$ where it is continuous, obeying the delay-differential equations

$$\frac{d}{ds}(sF(s)) = f(s-1) \tag{8}$$

for $s > 1$ and

$$\frac{d}{ds}(sf(s)) = F(s-1) \tag{9}$$

for $s > 2$, with the initial conditions

$$F(s) = \frac{2e^\gamma}{s} 1_{s>1}$$

for $s \leq 3$ and

$$f(s) = 0$$

for $s \leq 2$. Show that these properties of F, f determine F, f completely.

For future reference, we record the following explicit values of F, f:

$$F(s) = \frac{2e^\gamma}{s} \tag{10}$$

for $1 < s \leq 3$, and

$$f(s) = \frac{2e^\gamma}{s} \log(s-1) \tag{11}$$

for $2 \leq s \leq 4$.

We will show

Theorem 2 (Linear sieve) Let the notation and hypotheses be as above, with $s > 1$. Then, for any $\varepsilon > 0$, one has the upper bound

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n \leq (F(s) + O(\varepsilon)) XV(z) + O(\sum_{d \leq D: \mu^2(d)=1} |r_d|) \tag{12}$$

and the lower bound

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n \geq (f(s) - O(\varepsilon)) XV(z) + O(\sum_{d \leq D: \mu^2(d)=1} |r_d|) \tag{13}$$

if D is sufficiently large depending on ε, s, c. Furthermore, this claim is sharp in the sense that the quantity $F(s)$ cannot be replaced by any smaller quantity, and similarly $f(s)$ cannot be replaced by any larger quantity.

Comparing the linear sieve with the fundamental lemma (and also testing using the sequence $a_n = 1_{1 \leq n \leq N}$ for some extremely large N), we conclude that we necessarily have the asymptotics

$$1 - O(e^{-s}) \leq f(s) \leq 1 \leq F(s) \leq 1 + O(e^{-s})$$

for all $s \geq 1$; this can also be proven directly from the definitions of F, f, or from Exercise 1, but is somewhat challenging to do so; see e.g. Chapter 11 of Friedlander-Iwaniec for details.

> **Exercise 3** Establish the integral identities
>
> $$F(s) = 1 + \frac{1}{s}\int_s^\infty (1 - f(t-1))\ dt$$
>
> and
>
> $$f(s) = 1 + \frac{1}{s}\int_s^\infty (1 - F(t-1))\ dt$$
>
> for $s \geq 2$. Argue heuristically that these identities are consistent with the bounds in Theorem 2 and the Buchstab identity (Equation (16) from Notes 4).

> **Exercise 4** Use the Selberg sieve (Theorem 30 from Notes 4) to obtain a slightly weaker version of (12) in the range $1 < s < 3$ in which the error term $|r_d|$ is worsened to $\tau_3(d)|r_d|$, but the main term is unchanged.

We will prove Theorem 2 below the fold. The optimality of F, f is closely related to the parity problem obstruction discussed in Section 5 of Notes 4; a naive application of the parity arguments there only give the weak bounds $F(s) \geq \frac{2e^\gamma}{s}$ and $f(s) = 0$ for $s \leq 2$, but this can be sharpened by a more careful counting of various sums involving the Liouville function λ.

As an application of the linear sieve (specialised to the ranges in (10), (11)), we will establish a famous theorem of Chen, giving (in some sense) the closest approach to the twin prime conjecture that one can hope to achieve by sieve-theoretic methods:

> **Theorem 5 (Chen's theorem)** There are infinitely many primes p such that $p+2$ is the product of at most two primes.

The same argument gives the version of Chen's theorem for the even Goldbach conjecture, namely that for all sufficiently large even N, there exists a prime p between 2 and N such that $N-p$ is the product of at most two primes.

The discussion in these notes loosely follows that of Friedlander-Iwaniec (who study sieving problems in more general dimension than $\kappa = 1$).

— 1. Optimality —

We first establish that the quantities $F(s), f(s)$ appearing in Theorem 2 cannot be improved. We use the parity argument of Selberg, based on weight sequences a_n related to the Liouville function.

We argue for the optimality of $F(s)$; the argument for $f(s)$ is similar and is left as an exercise. Suppose that there is $s \geq 1$ for which the claim in Theorem 2 is not optimal, thus there exists $\delta > 0$ such that

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n \leq (F(s) - \delta) X V(z) + O\Big(\sum_{d \leq D : \mu^2(d) = 1} |r_d|\Big) \tag{14}$$

for $z, D, g, E_p, a_n, X, V(z), r_d$ as in that theorem, with z sufficiently large.

We will contradict this claim by specialising to a special case. Let z be a large parameter going to infinity, and set $D := z^s$. We set $g(d) := 1/d$, then by Mertens' theorem we have $V(z) = \frac{e^{-\gamma}+o(1)}{\log z}$. We set E_p to be the residue class $0\ (p)$, thus (3) becomes

$$\sum_{n : d \mid n} a_n = g(d) X + r_d \tag{15}$$

and (14) becomes

$$\sum_{n:(n,P(z))=1} a_n \leq \frac{F(s)-\delta+o(1)}{e^\gamma}\frac{X}{\log z} + O(\sum_{d\leq D:\mu^2(d)=1} |r_d|) \tag{16}$$

where $P(z) := \prod_{p<z} p$.

Now let $\varepsilon > 0$ be a small fixed quantity to be chosen later, set $X := D^{1+\varepsilon}$, and let a_n be the sequence

$$a_n := (1-\lambda(n))1_{1\leq n\leq X}.$$

This is clearly finitely supported and non-negative. For any d, we have

$$\sum_{n:d\mid n} a_n = \sum_{n\leq X/d} 1-\lambda(d)\sum_{n\leq X/d}\lambda(n)$$

from the multiplicativity of λ. If $d \leq D$, then $X/d \geq D^\varepsilon$, and then by the prime number theorem for the Liouville function (Exercise 41 from Notes 2, combined with Exercise 18 from Supplement 4) we have

$$\sum_{n\leq x/d}\lambda(n) \ll_\varepsilon \frac{X}{d}\log^{-10} D$$

(say), annd hence the remainder term r_d in (15) is of size

$$|r_d| \ll_\varepsilon \frac{X}{d}\log^{-10} D. \tag{17}$$

As such, the error term $O(\sum_{d\leq D:\mu^2(d)=1}|r_d|)$ in (16) may be absorbed into the $o(1)$ term, and so

$$\sum_{n\leq X:(n,P(z))=1}(1-\lambda(n)) \leq \frac{F(s)-\delta+o(1)}{e^\gamma}\frac{X}{\log z}. \tag{18}$$

Now we count the left-hand side. Observe that $1-\lambda(n)$ is supported on those numbers $n = p_1\dots p_r$ that are the product of an odd number of primes $p_1 \geq \dots \geq p_r$ (possibly with repetition), in which case $1-\lambda(n) = 2$. To be coprime to $P(z)$, all these primes must be at least z; since we are restricting $n \leq X = z^{(1+\varepsilon)s}$, we thus must have $r \leq (1+\varepsilon)s$. The left-hand side of (18) may thus be written as

$$2\sum_{r\leq(1+\varepsilon)s,\ \text{odd}}\ \sum_{p_1\geq\dots\geq p_r\geq z:p_1\dots p_r\leq X} 1. \tag{19}$$

This expression may be computed using the prime number theorem:

> **Exercise 6** Show that the expression (19) is equal to $\frac{F(s(1+\varepsilon))+o(1)}{e^\gamma}\frac{X}{\log z}$.

Since F is continuous for $s > 1$, we obtain a contradiction if ε is sufficiently small.

> **Exercise 7** Verify the optimality of $f(s)$ in Theorem 2. (*Hint:* replace $1-\lambda(n)$ by $1+\lambda(n)$ in the above arguments.)

— 2. The linear sieve —

We now prove the forward direction of Theorem 2. Again, we focus on the upper bound (12), as the lower bound case is similar.

Fix $s > 1$. Morally speaking, the most natural sieve to use here is the (upper bound) *beta sieve* from Notes 4, with the optimal value of β, which for the linear sieve turns out to be $\beta = 2$. Recall that this sieve is defined as the sum

$$\sum_{d\in\mathcal{D}_+}\mu(d)1_{E_d}$$

where $\mathcal{D}_+$ is the set of divisors $d = p_1\dots p_m$ of $P(z)$ with $z > p_1 > \dots > p_m$, such that

$$p_1\dots p_{r-1}p_r^3 \leq D$$

for all odd $1 \leq r \leq m$. From Proposition 14 of Notes 4 this is indeed an upper bound sieve; indeed we have

$$\sum_{d \in \mathcal{D}_+} \mu(d) 1_{E_d}(n) = 1_{n \notin \bigcup_{p<z} E_p} + \sum_{r \text{ odd}} \sum_{d \in \mathcal{E}_r} 1_{n \in E_d} 1_{n \notin \bigcup_{p<p_*(d)} E_p} \tag{20}$$

where $\mathcal{E}_r$ is the set of divisors $d = p_1 \ldots p_r$ of $P(z)$ with $z > p_1 > \ldots > p_r = p_*(d)$, such that

$$p_1 \ldots p_{r-1} p_r^3 > D \tag{21}$$

and

$$p_1 \ldots p_{r'-1} p_{r'}^3 \leq D \tag{22}$$

for all odd $1 \leq r' < r$. Now for the key heuristic point: if $n \approx D$ lies in the support of $1 - \lambda$, then the sum in (20) mostly vanishes. Indeed, if $n \leq D$ is such that $n \in E_d$ and $n \notin \bigcup_{p<p_*(d)} E_p$ for some $d = p_1 \ldots p_r \in \mathcal{E}_r$ and odd r, then one has $n = p_1 \ldots p_r q$ for some q that is not divisible by any prime less than p_r. On the other hand, from (21), (22) one has

$$q \approx \frac{D}{p_1 \ldots p_r} < p_r^2$$

and

$$q \approx \frac{D}{p_1 \ldots p_r} \geq 1$$

which (morally) implies from the sieve of Erathosthenes that q is prime, thus $\lambda(n) = (-1)^{r+1} = +1$ and so n is not in the support of $1 - \lambda$. As such, we expect the upper bound sieve $\sum_{d \in \mathcal{D}_+} \mu(d) 1_{E_d}$ to be extremely efficient on the support of $1 - \lambda$, which when combined with the analysis of the previous section suggests that this sieve should produce the desired upper bound (12).

One can indeed use this sieve to establish the required bound (12); see Chapter 11 of Friedlander-Iwaniec for details. However, for various technical reasons it will be convenient to modify this sieve slightly, by increasing the β parameter to be slightly greater than 2, and also by using the fundamental lemma to perform a preliminary sifting on the small primes.

We turn to the details. To prove (12) we may of course assume that $\varepsilon > 0$ is suitably small. It will be convenient to worsen the error $O(\varepsilon)$ in (12) a little to $O(\varepsilon \log \frac{1}{\varepsilon})$, since one can of course remove the logarithm by reducing ε appropriately.

Set $D_0 := z^{\varepsilon^2}$ and $z_0 := z^{\varepsilon^3}$. By the fundamental lemma of sieve theory (Lemma 17 of Notes 4), one can find combinatorial upper and lower bound sieve coefficients $(\lambda_d^{\pm,0})_{d \leq D_0}$ at sifting level z_0 supported on divisors of $P(z_0)$, such that

$$\sum_{d \leq D_0 : d \mid P(z_0)} \lambda_d^{\pm,0} g(d) = V(z_0)(1 + O(e^{-1/\varepsilon})). \tag{23}$$

Thus we have $\lambda_d^{\pm,0} \in \{-1, 0, 1\}$ and

$$\sum_{d \leq D_0 : d \mid P(z_0)} \lambda_d^{-,0} 1_{E_d}(n) \leq 1_{n \notin \bigcup_{p<z_0} E_p} \leq \sum_{d \leq D_0 : d \mid P(z_0)} \lambda_d^{+,0} 1_{E_d}(n) \tag{24}$$

for all n.

We will use the upper bound sieve $\sum_{d \leq D_0 : d \mid P(z_0)} \lambda_d^{+,0} 1_{E_d}(n)$ as a preliminary sieve to remove the E_p for $p < z_0$; the lower bound sieve $\lambda_d^{-,0}$ plays only a minor supporting role, mainly to control the difference between the upper bound sieve $1_{n \notin \bigcup_{p<z_0} E_p}$.

Next, we set $P(z_0,z) := P(z)/P(z_0) = \prod_{z_0 \leq p < z} p$, and let λ_d^+ be the upper bound beta sieve with parameter $\beta = 2+\varepsilon$ on the primes dividing $P(z_0,z)$ up to level of distribution D/D_0^2. In other words, $\mathcal{D}_+$ consists of those divisors $d = p_1 \dots p_r$ of $P(z_0,z)$ with $p_1 > \dots > p_r$ such that

$$p_1 \dots p_{m-1} p_m^{3+\varepsilon} \leq D/D_0^2$$

for all odd $1 \leq m \leq r$; in particular, $d \leq D/D_0^2$ for all $d \in \mathcal{D}_+$. By Proposition 14 of Notes 4, this is indeed an upper bound sieve for the primes dividing $P(z_0,z)$:

$$\sum_{d \in \mathcal{D}_+} \mu(d) 1_{E_d} \geq 1_{n \notin \bigcup_{z_0 \leq p < z} E_p}.$$

Multiplying this with the second inequality in (24) (this is the method of *composition of sieves*), we obtain an upper bound sieve for the primes up to z:

$$\sum_{d_0 \leq D_0 : d_0 | P(z_0)} \sum_{d \in \mathcal{D}_+} \lambda_{d_0}^{+,0} \mu(d) 1_{E_{d_0 d}}(n) \geq 1_{n \notin \bigcup_{p<z} E_p}.$$

Multiplying this by a_n and summing in n, we conclude that

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n \leq \sum_{d_0 \leq D_0 : d_0 | P(z_0)} \sum_{d \in \mathcal{D}_+} \lambda_{d_0}^{+,0} \mu(d) \sum_{n \in E_{d_0 d}} a_n.$$

Note that each product $d_0 d$ appears at most once in the above sum, and all such products are squarefree and at most D. Applying (3), we thus have

$$\sum_{n \notin \bigcup_{p<z} E_p} a_n \leq \sum_{d_0 \leq D_0 : d_0 | P(z_0)} \sum_{d \in \mathcal{D}_+} \lambda_{d_0}^{+,0} \mu(d) g(d_0) g(d) X + \sum_{d \leq D : \mu^2(d) = 1} |r_d|.$$

Thus to prove (12), it suffices to show that

$$\sum_{d_0 \leq D_0 : d_0 | P(z_0)} \sum_{d \in \mathcal{D}_+} \lambda_{d_0}^{+,0} \mu(d) g(d_0) g(d) = (F(s) + O(\varepsilon \log \frac{1}{\varepsilon})) V(z)$$

for z sufficiently large depending on ε. Factoring out the d_0 summation using (23), it thus suffices to show that

$$\sum_{d \in \mathcal{D}_+} \mu(d) g(d) = (F(s) + O(\varepsilon \log \frac{1}{\varepsilon})) V(z)/V(z_0).$$

Now we eliminate the role of g. From (1), (5) and Mertens' theorem we have

$$V(z)/V(z_0) = (1 + O(\varepsilon)) \frac{\log z_0}{\log z}$$

for z large enough. Also, if $d \in \mathcal{D}_+$, then $d \leq D/D_0^2$ and all prime factors of d are at least $z_0 = D^{\varepsilon^3/s}$. Thus d has at most $O_{s,\varepsilon}(1)$ prime factors, each of which are at least z_0. From (1) we then have

$$g(d) = \frac{1}{d} + O_{s,\varepsilon}(\frac{1}{z_0} \frac{1}{d}).$$

The contribution of the error term $O_{s,\varepsilon}(\frac{1}{z_0}\frac{1}{d})$ is then easily seen to be $O(\varepsilon \frac{\log z_0}{\log z})$ for z large enough, and so we reduce to showing that

$$\sum_{d \in \mathcal{D}_+} \frac{\mu(d)}{d} = (F(s) + O(\varepsilon \log \frac{1}{\varepsilon})) \frac{\log z_0}{\log z}. \tag{25}$$

One can proceed here by evaluating the left-hand side directly. However, we will proceed instead by using the weight function $1 - \lambda$ from the previous section. More precisely, we will evaluate the expression

$$\sum_{n\leq D}(1-\lambda(n))\sum_{d_0\leq D_0:d_0|\ P(z_0)}\sum_{d\in\mathcal{D}_+}\lambda^{+,0}_{d_0}\mu(d)1_{d_0d\mid n} \tag{26}$$

in two different ways, where $\lambda^{+,0}_{d_0}$ is as before (but with the role of $g(d)$ now replaced by the function $1/d$). Firstly, since $d_0d\leq D/D_0$, we see from the argument used to establish (17) that

$$\sum_{n\leq D:d_0d\mid n}1-\lambda(n)=\frac{D}{d_0d}+O(\frac{D}{d_0d}\log^{-10}D)$$

(say). Since each d_0d appears at least once, we can thus write (26) as

$$\sum_{d_0\leq D_0:d_0|\ P(z_0)}\sum_{d\in\mathcal{D}_+}\lambda^{+,0}_{d_0}\mu(d)\frac{D}{d_0d}+O(D\log^{-9}D)$$

which upon factoring the d_0 sum using (23) and Mertens' theorem

$$(1+O(\varepsilon))\frac{D}{e^\gamma\log z_0}\sum_{d\in\mathcal{D}_+}\frac{\mu(d)}{d}+O(D\log^{-9}D).$$

Thus to verify (25), it will suffice to show that (26) is of the form

$$(F(s)+O(\varepsilon\log\frac{1}{\varepsilon}))\frac{D}{e^\gamma\log z}$$

for z sufficiently large.

To do this, we abbreviate (26) as

$$\sum_{n\leq D}(1-\lambda(n))\nu^{+,0}(n)\sum_{d\in\mathcal{D}_+}\mu(d)1_{d\mid n} \tag{27}$$

where $\nu^{\pm,0}$ are the sieves

$$\nu^{\pm,0}(n):=\sum_{d_0\leq D_0:d_0|\ P(z_0)}\lambda^{\pm,0}_{d_0}1_{d_0|\ n}.$$

By Proposition 14 of Notes 4, we can expand $\sum_{d\in\mathcal{D}_+}\mu(d)1_{d\mid n}$ as

$$1_{(n,P(z_0,z))=1}+\sum_{r\text{ odd}}\sum_{d\in\mathcal{E}_r}1_{d\mid n}1_{(n,P(z_0,p_*(d)))=1} \tag{28}$$

where for any $r\geq 1$, $\mathcal{E}_r$ is the collection of all divisors $d=p_1\ldots p_r$ of $P(z_0,z)$ with $p_1>\ldots>p_r=p_*(d)$ such that

$$p_1\ldots p_{r-1}p_r^{3+\varepsilon}>D/D_0^2 \tag{29}$$

but

$$p_1\ldots p_{r'-1}p_{r'}^{3+\varepsilon}\leq D/D_0^2 \tag{30}$$

for all $1\leq r'<r$ with the same parity as r. For technical reasons, we will also impose the additional inequality

$$p_1\ldots p_{r'-1}p_{r'}^{2+\varepsilon}\leq D/D_0^2 \tag{31}$$

for all $1\leq r'<r$ with the opposite parity as r; this follows from (30) when $r'>1$, but is an additional constraint when $r'=1$ and r is even, but in the above identity r is odd, so this additional constraint is harmless. For similar reasons we impose the inequality

$$p_1\ldots p_r^{1+\varepsilon}\leq D/D_0^2 \tag{32}$$

which follows from (30) or (31) except when $r=1$, but then this inequality is automatic from our hypothesis $s>1$, which implies $z^{1+\varepsilon}\leq D/D_0^2$ if ε is chosen small enough.

Inserting the identity (28), we can write (26) as

$$A + \sum_{r \text{ odd}} B_r$$

where

$$A := \sum_{n \leq D} (1 - \lambda(n)) \nu^{+,0}(n) 1_{(n, P(z_0, z))=1}$$

and

$$B_r := \sum_{d \in \mathcal{E}_r} \sum_{n \leq D} (1 - \lambda(n)) \nu^{+,0}(n) 1_{d \mid n} 1_{(n, P(z_0, p_*(d))=1}.$$

We first estimate A. By(24), we can write A as the sum of

$$\sum_{n \leq D} (1 - \lambda(n)) 1_{(n, P(z))=1} \tag{33}$$

plus an error of size at most

$$O\Big(\Big| \sum_{n \leq D} \Big(\sum_{d_0 \leq D_0 : d_0 \mid P(z_0)} \lambda^{+,0}_{d_0} 1_{d_0 \mid n} - \sum_{d_0 \leq D_0 : d_0 \mid P(z_0)} \lambda^{-,0}_{d_0} 1_{d_0 \mid n} \Big) \Big| \Big)$$

(where we bound $(1 - \lambda(n)) 1_{(n, P(z_0, z))=1}$ by $O(1)$). The error may be rearranged as

$$O\Big(\Big| \sum_{d_0 \leq D_0 : d_0 \mid P(z_0)} (\lambda^{+,0}_{d_0} - \lambda^{-,0}_{d_0}) \Big(\frac{D}{d_0} + O(1) \Big) \Big| \Big),$$

which by (23) is of size $O(e^{-1/\varepsilon} \frac{D}{\log z_0}) = O(\varepsilon \frac{D}{\log z})$ for ε small enough. As for the main term (33), we see from Exercise 6 (and the arguments preceding that exercise) that this term is equal to $\frac{F(s) + O(\varepsilon)}{e^\gamma} \frac{D}{\log z}$ for z sufficiently large. Thus, to obtain the desired approximation for (26), it will suffice to show that

$$\sum_{r \text{ odd}} B_r \ll \varepsilon \log \frac{1}{\varepsilon} \frac{D}{\log z}.$$

Next, we establish an exponential decay estimate on the B_r:

> **Lemma 8** For z sufficiently large depending on ε, we have
>
> $$B_r \ll \varepsilon^{-O(1)} \alpha^r e^{-s} \frac{D}{\log z}$$
>
> for all $r \geq 1$ and some absolute constant $0 < \alpha < 1$.

Proof: (Sketch) Note that if $d = p_1 \dots p_r$ is in $\mathcal{E}_r$, then $d \leq D/D_0^2$ and all prime factors are at least $z_0 = D^{\varepsilon^3}$, thus we may assume without loss of generality that $r \leq 1/\varepsilon^3$.

We bound

$$B_r \ll \sum_{d \in \mathcal{E}_r} \sum_{n \leq D/d} \nu^{+,0}(n) 1_{(n, P(z_0, p_*(d)))=1}.$$

Note that if $d = p_1 \dots p_r$ lies in $\mathcal{E}_r$, then

$$D_0^2 p_r^\varepsilon \leq \frac{D}{d} < D_0^2 p_r^{2+\varepsilon}$$

thanks to (29), (32). From this and the fundamental lemma of sieve theory we see (Exercise!) that

$$\sum_{n \leq D/d} \nu^{+,0}(n) 1_{(n, P(z_0, p_r))=1} \ll \varepsilon^{-O(1)} \frac{D}{d \log p_*(d)}$$

and so it will suffice to show that

$$\sum_{d \in \mathcal{E}_r} \frac{1}{d} \frac{\log z}{\log p_*(d)} \ll \alpha^r e^{-s}. \tag{34}$$

By the prime number theorem, the left-hand side is bounded (Exercise!) by $f_r(s - 2\varepsilon^2) + o(1)$ as $z \to \infty$, where

$$f_r(s) := \int_{J_r(s)} \frac{dt_1 ... dt_r}{t_1 ... t_{r-1} t_r^2}$$

and $J_r(s) \subset (0, +\infty)^r$ is the set of points $(t_1, ..., t_r)$ with $1 \geq t_1 \geq ... \geq t_r > 0$,

$$t_1 + ... + t_{r-1} + (3+\varepsilon) t_r \geq s,$$

and such that

$$t_1 + ... + t_{r'-1} + (3+\varepsilon) t_{r'} \leq s$$

for all $1 \leq r' < r$ with the same parity as r, and

$$t_1 + ... + t_{r'-1} + (2+\varepsilon) t_{r'} \leq s$$

for all $1 \leq r' \leq r$. It thus suffices to prove the bound

$$f_r(s) \leq C \alpha^r e^{-s} \tag{35}$$

for all $s > 1$ and some absolute constant $C > 0$.

We use an argument from the book of Harman. Observe that $f_r(s)$ vanishes for $s > r + 2$, which makes the claim (35) routine for $r = 1$ (Exercise!) for C sufficiently large. We will now inductively prove (35) for all odd r. From the change of variables $u = \frac{s}{t_1}$, we obtain the identity

$$f_r(s) = \frac{1}{s} \int_{\max(s, b_r)}^{\infty} f_{r-1}(t-1)\, dt \tag{36}$$

where $b_r := 3 + \varepsilon$ when r is odd and $b_r := 2 + \varepsilon$ when r is even (Exercise!). In particular, if $r \geq 3$ is odd and (35) was already proven for $r - 2$, then

$$\begin{aligned} f_r(s) &= \frac{1}{s} \int_{\max(s, 3+\varepsilon)}^{\infty} \frac{1}{t-1} \int_{t-1}^{\infty} f_{r-2}(u-1)\, du dt \\ &\leq \frac{1}{s} \int_{\max(s, 3+\varepsilon)}^{\infty} \frac{1}{t-1} \int_{t-1}^{\infty} C \alpha^{r-2} e^{1-u}\, du dt \\ &\leq C \alpha^r e^{-s} \times \alpha^{-2} \times \frac{e^{s+2}}{s} \int_{\max(s, 3+\varepsilon)}^{\infty} \frac{e^{-t}}{t-1}\, dt. \end{aligned}$$

One can check (Exercise!) that the quantity $\frac{e^{s+2}}{s} \int_{\max(s, 3+\varepsilon)}^{\infty} \frac{e^{-t}}{t-1}\, dt$ is maximised at $s = 3 + \varepsilon$, where its value is less than 1 (in fact it is 0.94...) if ε is small enough. As such, we obtain (35) if α is sufficiently close to 1.

Finally, (35) for even r follows from the odd r case (with a slightly larger choice of C) by one final application of (36). □

> **Exercise 9** Fill in the steps marked (Exercise!) in the above proof.

In view of this lemma, the total contribution of B_r with $r > C \log \frac{1}{\varepsilon}$ for some sufficiently large C is acceptable. Thus it suffices to show that

$$B_r \ll \varepsilon \frac{D}{\log z}$$

whenever $r = O(\log \frac{1}{\varepsilon})$ is odd.

By (24), we can write B_r as

$$\sum_{d\in\mathcal{E}_r}\sum_{n\leq D}(1-\lambda(n))1_{d\mid n}1_{(n,P(p_*(d)))=1}$$

plus an error of size

$$O(\sum_{d\in\mathcal{E}_r}\sum_{n\leq D}(\sum_{d_0\leq D_0:d_0\mid P(z_0)}\lambda_{d_0}^{+,0}1_{dd_0\mid n}-\sum_{d_0\leq D_0:d_0\mid P(z_0)}\lambda_{d_0}^{-,0}1_{dd_0\mid n})).$$

Arguing as in the treatment of the A term, we see from (23) that the error term is bounded by

$$\begin{aligned}
&\ll O(\sum_{d\in\mathcal{E}_r}\sum_{d_0\leq D_0:d_0\mid P(z_0)}(\lambda_{d_0}^{+,0}-\lambda_{d_0}^{-,0})\frac{D}{dd_0})+\sum_{d\in\mathcal{E}_r}\sum_{d_0\leq D_0:d_0\mid P(z_0)}1\\
&\ll \sum_{d\in\mathcal{E}_r}e^{-1/\varepsilon}\frac{D}{d\log z_0}+\frac{D}{D_0^2}D_0\\
&\ll e^{-1/\varepsilon}\frac{D}{\log z_0}(\sum_{z_0\leq p\leq z}\frac{1}{p})^r+\frac{D}{D_0}\\
&\ll e^{-1/\varepsilon}\frac{D}{\log z_0}O(\log\frac{\log z}{\log z_0})^r+\frac{D}{D_0}\\
&\ll \varepsilon\frac{D}{\log z}
\end{aligned}$$

as desired for z large enough, since $r=O(\log\frac{1}{\varepsilon})$ and $\frac{\log z}{\log z_0}=O(\varepsilon^{-3})$. Thus it suffices to show that

$$\sum_{d\in\mathcal{E}_r}\sum_{n\leq D}(1-\lambda(n))1_{d\mid n}1_{(n,P(p_*(d)))=1}\ll\varepsilon\frac{D}{\log z}. \tag{37}$$

If $d=p_1\dots p_r$ and n appear in the above sum, then we have $n=p_1\dots p_rq$ where $q\leq\frac{D}{d}$ has no prime factor less than p_r, has an even number of prime factors, and obeys the bounds

$$q\leq D_0^2p_r^{2+\varepsilon}$$

thanks to (29). Note that (29) also gives $p_r\geq(D/D_0^2)^{1/(r+2)}$, and thus (since $r=O(\log\frac{1}{\varepsilon})$ and $D_0=z^{\varepsilon^3}$) we see that $q<p_r^3$ if ε is small enough and z is large enough. This forces q to either equal 1, or be the product of two primes between p_r and $D_0^2p_r^{1+\varepsilon}$. The contribution of the $q=1$ case is bounded by $|\mathcal{E}_r|\leq D/D_0^2$, which is acceptable. As for the contribution of those q that are the product of two primes, the prime number theorem shows that there are at most

$$\sum_{p_r\leq p\leq D_0^2p_r^{1+\varepsilon}}O(\frac{D}{dp\log p_r})=O(\varepsilon\frac{D}{d\log p_r})$$

values of q that can contribute to the sum, and so this contribution to B_r is at most

$$O(\varepsilon\frac{D}{\log z}\sum_{d\in\mathcal{E}_r}\frac{1}{d}\frac{\log z}{\log p_*(d)}),$$

but by (34) the sum here is $O(1)$ for z large enough, and the claim follows. This completes the proof of (12).

> **Exercise 10** Establish the lower bound (13) in Theorem 2. (Note that one can assume without loss of generality that $s>2$, which will now be needed to ensure (31) when $r'=1$.)

— 3. Chen's theorem —

We now prove Chen's theorem for twin primes, loosely following the treatment in Chapter 25 of Friedlander-Iwaniec. We will in fact show the slightly stronger statement that

$$\sum_{x/2 \leq n \leq x-2} \Lambda(n) 1_{\mathcal{P}_2}(n+2) 1_{(n+2,P(z))=1} \gg \frac{x}{\log x}$$

for sufficiently large x, where $\mathcal{P}_2$ is the set of all numbers that are products of at most two primes, and $z := x^{1/8}$. Indeed, after removing the (negligible) contribution of those n that are powers of primes, this estimate would imply that there are infinitely many primes p such that $p+2$ is the product of at most two primes, each of which is at least $p^{1/8}$.

Chen's argument begins with the following simple lower bound sieve for $1_{\mathcal{P}_2}$:

Lemma 11 If $x^{2/3} < n \leq x$, then

$$1_{\mathcal{P}_2}(n) \geq 1 - \frac{1}{2} \sum_{p \leq x^{1/3}} 1_{p \mid n} - \frac{1}{2} \sum_{p_1 \leq x^{1/3} < p_2 \leq p_3} 1_{n=p_1 p_2 p_3} - \frac{1}{2} \sum_{p \leq x^{1/3}} 1_{p^2 \mid n}.$$

Proof: If n has no prime factors less than or equal to $x^{1/3}$, then $n \in \mathcal{P}_2$ and the claim follows. If n has two or more factors less than or equal to $x^{1/3}$, then $1 - \frac{1}{2}\sum_{p \leq x^{1/3}} 1_{p \mid n} \leq 0$ and the claim follows. Finally, if n has exactly one factor less than or equal to $x^{1/3}$, then (as $n > x^{2/3}$) it must be of the form $n = p_1 p_2 p_3$ for some $p_1 \leq x^{1/3} < p_2 \leq p_3$, or it is divisible by p^2 for some $p \leq x^{1/3}$, and the claim again follows. □

In view of this sieve (trivially managing the contribution of $n \leq x^{2/3}$ or of the case where $p^2 \mid n$, and using the restriction of $n+2$ to be coprime to $P(z)$), it suffices to show that

$$A_1 - \frac{1}{2} \sum_{z \leq p \leq x^{1/3}} A_{2,p} - \frac{1}{2} A_3 \gg \frac{x}{\log x} \tag{38}$$

for sufficiently large x, where

$$A_1 := \sum_{x/2 \leq n \leq x-2} \Lambda(n) 1_{(n+2,P(z))=1},$$

$$A_{2,p} := \sum_{x/2 \leq n \leq x-2 : p \mid n+2} \Lambda(n) 1_{(n+2,P(z))=1},$$

and

$$A_3 := \sum_{x/2 \leq n \leq x-2} \Lambda(n) \sum_{z \leq p_1 \leq x^{1/3} < p_2 \leq p_3} 1_{n+2=p_1 p_2 p_3}.$$

We thus seek sufficiently good lower bounds on A_1 and sufficiently good upper bounds on $A_{2,p}$ and A_3. As it turns out, the linear sieve, combined with the Bombieri-Vinogradov theorem, will give bounds on $A_1, A_{2,p}, A_3$ with numerical constants that are sufficient for this purpose.

We begin with A_1. We use the lower bound linear sieve, with E_p equal to the residue class -2 (p) for all $p < z$, so that E_d is the residue class -2 (d). We approximate

$$\sum_{x/2 \leq n \leq x-2 : n \in E_d} \Lambda(n) = g(d) \frac{x}{2} + r_d$$

where g is the multiplicative function with $g(2) := 0$ and $g(p) := \frac{1}{p-1}$ for $p > 2$. From the Bombieri-Vinogradov theorem (Theorem 17 of Notes 3) we have

$$\sum_{d \leq D} |r_d| \ll x \log^{-10} x \tag{39}$$

(say) if $D := x^{1/2-\varepsilon} = z^{4-8\varepsilon}$ for some small fixed $\varepsilon > 0$. Applying the lower bound linear sieve (13), we conclude that

$$A_1 \geq (f(4-8\varepsilon) - O(\varepsilon)) \frac{x}{2} V(z) + O(x \log^{-10} x)$$

where

$$V(z) = \prod_{p<z}(1-g(p)).$$

We can compute an asymptotic for V:

> **Exercise 12** Show that
>
> $$V(z) = (2\Pi_2 + o(1))\frac{1}{e^\gamma \log z}$$
>
> as $z \to \infty$, where $\Pi_2 = \prod_{p>2}(1-\frac{1}{(p-1)^2})$ is the twin prime constant.

From (11) we have $f(4) = \frac{e^\gamma}{2}\log 3$. Sending ε slowly to zero, we conclude that

$$A_1 \geq (\log 3 - o(1))\Pi_2 \frac{x}{2\log z}. \tag{40}$$

Now we turn to $A_{2,p}$. Here we use the upper bound linear sieve. Let E_d be as before. For any d dividing $P(z)$ and $z \leq p < x^{1/3}$, we have

$$\sum_{x/2 \leq n \leq x-2 : n \in E_d} \Lambda(n) 1_{p \mid n+2} = g(d)g(p)\frac{x}{2} + r_{pd}$$

where g and r_d are as previously. We apply the upper bound linear sieve (12) with level of distribution D/p, to conclude that

$$A_{2,p} \leq (F(\frac{\log D/p}{\log z}) + O(\varepsilon))g(p)\frac{x}{2}V(z)$$
$$+O(\sum_{d \leq D/p} |r_{pd}|).$$

We sum over p. Since pd is at most D, and each number less than or equal to D has at most $O(\log x)$ prime factors, we have

$$\sum_{z \leq p < x^{1/3}} A_{2,p} \leq \sum_{z \leq p < x^{1/3}} (F(\frac{\log D/p}{\log z})$$
$$+O(\varepsilon))g(p)\frac{x}{2}V(z) + O(\log x \sum_{d \leq D} |r_d|).$$

The error term is $O(x \log^{-9} x)$ thanks to (39). Since $g(p) = \frac{1+o(1)}{p}$, we thus have

$$\sum_{z \leq p < x^{1/3}} A_{2,p} \leq (\sum_{z \leq p < x^{1/3}} \frac{F(\frac{\log D/p}{\log z})}{p} + O(\varepsilon))\frac{\Pi_2 x}{e^\gamma \log z}$$

for sufficiently large x, thanks to Exercise 12. We can compute the sum using Exercise 37 of Notes 1, to obtain

$$\sum_{z \leq p < x^{1/3}} \frac{F(\frac{\log D/p}{\log z})}{p} = \int_1^{8/3} F(4-8\varepsilon-t)\frac{dt}{t} + o(1),$$

which by (10) and sending $\varepsilon \to 0$ slowly gives

$$\sum_{z \leq p < x^{1/3}} A_{2,p} \leq (\int_1^{8/3} \frac{1}{4-t}\frac{dt}{t} + o(1))\frac{2\Pi_2 x}{\log z}.$$

A routine computation shows that

$$\int_1^{8/3} \frac{1}{4-t}\frac{dt}{t} = \frac{\log 6}{4}$$

and so

$$\sum_{z\leq p<x^{1/3}} A_{2,p} \leq (\log 6 + o(1))\Pi_2 \frac{x}{2\log z}. \tag{41}$$

Finally, we consider A_3, which is estimated by "switching" the sieve to sift out small divisors of n, rather than small divisors of $n+2$. Removing those n with $n \leq \sqrt{x}$, as well as those n that are powers of primes, and then shifting n by 2, we have

$$A_3 \leq \log x \sum_n 1_{(n-2,P(\sqrt{x}))=1} a_n + O_\varepsilon(x^{1/2+\varepsilon})$$

where a_n is the finitely supported non-negative sequence

$$a_n := 1_{x/2+2\leq n\leq x} \sum_{z\leq p_1\leq x^{1/3}<p_2\leq p_3} 1_{n=p_1p_2p_3}. \tag{42}$$

Here we are sifting out the residue classes $E'_p := 2\ (p)$, so that $E'_d := 2\ (d)$.

The sequence a_n has good distribution up to level D:

Proposition 13 One has

$$\sum_{n\in E'_d} a_n = g(d) \sum_n a_n + r_d$$

where $g(d)$ is as before, and

$$\sum_{d\leq D:\mu^2(d)=1} |r_d| \ll x\log^{-10} x$$

(say), with D as before.

Proof: Observe that the quantity p_3 in (42) is bounded above by $x^{2/3}$ if the summand is to be non-zero. We now use a finer-than-dyadic decomposition trick similar to that used in the proof of the Bombieri-Vinogradov theorem in Notes 3 to approximate a_n as a combination of Dirichlet convolutions. Namely, we set $\lambda := 1+\log^{-20} x$, and partition $[x^{1/3}, x^{2/3}]$ (plus possibly a little portion to the right of $x^{2/3}$) into $A = O(\log^{21} x)$ consecutive intervals $I_1, \ldots, I_A$ each of the form $[N, \lambda N]$ for some $x^{1/3} \leq N \leq x^{2/3}$. We similarly split $[z, x^{1/3}]$ (plus possibly a tiny portion of $[z/2, z]$) into $B = O(\log^{21} x)$ intervals $J_1, \ldots, J_B$ each of the form $[M, \lambda M]$ for some $1 \ll M \leq x^{1/3}$. We can thus split a_n as

$$a_n = \sum_{1\leq b\leq B} \sum_{1\leq a\leq a'\leq A} \sum_{z\leq p_1\leq x^{1/3}<p_2\leq p_3: p_1\in J_b, p_2\in I_a, p_3\in I_{a'}} 1_{n=p_1p_2p_3} 1_{x/2+2\leq n\leq x}.$$

Observe that for each a, a' there are only $O(1)$ choices of b for which the summand can be non-zero. As such, the contribution of the diagonal case $a = a'$ can be easily seen to be absorbed into the r_d error, as can those cases where the product set $\{uvw : u \in J_b, v \in I_a, r \in I_{a'}\}$ is not contained completely in $[x/2+2, x]$. If we let Ω be the set of triplets (b, a, a') obeying these properties, we can thus approximate a_n by $\sum_{(b,a,a')\notin\Omega} a_n^{(b,a,a')}$, where $a_n^{(b,a,a')}$ is the Dirichlet convolution

$$a_n^{(b,a,a')} := 1_{\mathcal{P}\cap J_b} * 1_{\mathcal{P}\cap I_a} * 1_{\mathcal{P}\cap I_{a'}}.$$

From the general Bombieri-Vinogradov theorem (Theorem 16 of Notes 3) and the Siegel-Walfisz theorem (Exercise 64 of Notes 2) we see that

$$\sum_{n\in E_d} a_n^{(b,a,a')} = g(d) X^{(b,a,a')} + r_d^{(b,a,a')}$$

where

$$\sum_{d\leq D:\mu^2(d)=1} \left|r_d^{(b,a,a')}\right| \ll x\log^{-10} x$$

(say) and

$$X^{(b,a,a')} = \sum_n a_n^{(b,a,a')}.$$

This gives the claim with $\sum_n a_n$ replaced by the quantity $\sum_{(b,a,a') \notin \Omega} X^{(b,a,a')}$; but by undoing the previous decomposition we see that this quantity is equal to $\sum_n a_n$ up to an error of $O(x \log^{-11} x)$ (say), and the claim follows.

Applying the upper bound sieve (12) (with sifting level $D^{1/(1+\varepsilon)}$), we thus have

$$\sum_n 1_{(n-2,P(\sqrt{x}))=1} a_n \leq (F(1+\varepsilon) + O(\varepsilon)) V(D^{1/(1+\varepsilon)}) \sum_n a_n + O(x \log^{-10} x)$$

and hence by (10) and Exercise 12

$$\sum_n 1_{(n-2,P(\sqrt{x}))=1} a_n \leq (4 + O(\varepsilon)) \frac{2\Pi_2}{\log x} \sum_n a_n + O(x \log^{-10} x)$$

for x sufficiently large.

Note that

$$\sum_n a_n = \sum_{z \leq p_1 \leq x^{1/3} < p_2 < \frac{x}{p_1 p_2}} \sum_{\max(p_2, \frac{x/2+2}{p_1 p_2}) < p_3 < \frac{x}{p_1 p_2}} 1.$$

From the prime number theorem and Exercise 37 of Notes 1, we thus have

$$\sum_n a_n \leq (1+o(1)) \frac{x}{2 \log x} \int_{1/8 \leq t_1 \leq 1/3 < t_2 < 1-t_1-t_2} \frac{dt_1 dt_2}{t_1 t_2 (1-t_1-t_2)}.$$

(In fact one also has a matching lower bound, but we will not need it here.) We thus conclude that

$$A_3 \leq (C + o(1)) \Pi_2 \frac{x}{2 \log z}$$

where

$$\begin{aligned} C &:= \int_{1/8 \leq t_1 \leq 1/3 < t_2 < 1-t_1-t_2} \frac{dt_1 dt_2}{t_1 t_2 (1-t_1-t_2)} \\ &= 0.363\ldots \end{aligned}$$

The left-hand side of (38) is then at least

$$(2 \log 3 - \log 6 - C) \Pi_2 \frac{x}{4 \log z}.$$

One can calculate that $2 \log 3 - \log 6 - C > 0$, and the claim follows.

> **Exercise 14** Establish Chen's theorem for the even Goldbach conjecture.

> **Remark 15** If one is willing to use stronger distributional claims on the primes than is provided by the Bombieri-Vinogradov theorem, then one can use a simpler sieve than Chen's sieve to establish Chen's theorem, but then the required distributional theorem will then either be conjectural or more difficult to establish than the Bombieri-Vinogradov theorem. See Chapter 25 of Friedlander-Iwaniec for further discussion.

Comments (24)

Eytan Paldi · 29 January 2015

It seems that in the RHS of (6)-(7), "u" (in the integrands) should be "s".
[Corrected, thanks – T.]

> **Eytan Paldi** · 29 January 2015
>
> Also in the line below (7).
> *[Corrected, thanks – T.]*

arch1 · 29 January 2015

In the last two integrals preceding Exercise 9, the upper integration limit is misplaced.
[Corrected, thanks -T.]

arch1 · 29 January 2015

3 lines up from (37): we see from (23) *that the error*(?) is bounded by
[Corrected, thanks -T.]

Lior Silberman · 1 February 2015

In the proof of Chen's Theorem, in the paragraph "We begin with A_1", E_d is said to be the class of -2 mod p rather than mod d.
[Corrected, thanks – T.]

Karaskas · 31 March 2015

Just after the fragment "and the Siegel-Walfisz theorem (Exercise 64 of Notes 2) we see that [Formula] where" it should probably be log^{-10}x instead of log^{-100x}.
[Corrected, thanks – T.]

Karaskas · 31 March 2015

Right now it's log^{-10x} instead of log^{-10}x.
[Corrected, thanks – T.]

Karaskas · 10 April 2015

Chen's theorem section: shouldn't it be $x/2 \leq n \leq x-2 : p \mid n+2$ under the big sigma in the definition of $A_{2,p}$? The same thing reappears a few lines later.
Just after "Now we turn to $A_{2,p}$", what happens if p divides d?
[I was not able to locate this issue – T.]

> **Karaskas** · 11 April 2015
>
> In the same fragment it probably should be $z \leq p \leq x^{1/3}$ instead of $z^{1/3} \leq p \leq z$, shouldn't it? That would also explain my last question.
> *[Corrected, thanks – T.]*
>
> > **Karaskas** · 11 April 2015
> >
> > The $A_{2,p}$ issue wasn't there anymore while I was writting about it. I think you had repaired it a few days before and I forgot to refresh the website. Sorry for the confusion.
> > By the way, thank you for writing posts about sieve theory. Some details and ideas are much easier for me to follow right now.

Karaskas · 14 April 2015

The formula under "Applying the upper bound sieve" fragment: it should be $O(\varepsilon)$ instead of ε in one place.
[Corrected, thanks – T.]

Karaskas · 15 December 2015

I have a little confusion: before the equation (40) we send {\varepsilon} to 0 slowly but in Theorem 2 we have "if {D} is sufficiently large depending on {\varepsilon, s, c}." How should I understand this?

> **Karaskas** · 15 December 2015
>
> It seems to me like we are using two different epsilons there. By the way, is it possible to relax conditions in Theorem 2 and let ε depend on z?
> Theorem 2 also reminds me the Jurkat-Richert theorem, although we do not have any restriction on the sieve dimension. How do these two theorems are related to each other?
> Sorry for doubling the post and using LaTeX wrongly in the previous one.
>
> **Terence Tao** · 15 December 2015
>
> For any fixed choice of ε, the largeness condition on D required for Theorem 2 will apply if x is large enough, thus
> $A_1 \geq (f(4-8\varepsilon) - O(\varepsilon))\Pi_2 \frac{x}{\log z}$
> whenever x is sufficiently large depending on ε (this allows for the o(1) errors to be absorbed into the $O(\varepsilon)$ error). In particular, for any $\delta > 0$, one has
> $A_1 \geq (f(4) - \delta)\Pi_2 \frac{x}{\log z}$
> whenever x is sufficiently large depending on δ (by choosing ε appropriately depending on δ). This implies that
> $A_1 \geq (f(4) - o(1))\Pi_2 \frac{x}{\log z}$
> as $x \to \infty$.
> Another way to think about it as follows: if one has a bound of the form
> $f(x) = O(\varepsilon) + o(1) \quad (1)$
> as $x \to \infty$ for any fixed choice of ε, then one automatically has the improvement
> $f(x) = o(1)$
> as $x \to \infty$, by setting ε equal to a function of x that goes to zero sufficiently slowly (slow enough so that the $o(1)$ error in (1) still goes to zero even though ε is now varying).
> The Jurkat-Richert theorem has a slightly better error term replacing $O(\varepsilon)$, and a slightly worse error term for the $|r_d|$ errors, but can easily be used as a substitute for Theorem 2 for the purpose of proving Chen's theorem.

Karaskas · 30 December 2015

I have one problem with Lemma 11. How do we rule out the numbers of the form $p^2 q$, where $p, q \in \mathcal{P}$ and $p \leq x^{1/3} - 1$ and $q \geq x^{1/3}$?
I saw an another proof of Chen's theorem in which the author (Nathanson, I guess) used the sum $\sum_{z<q\leq x^{1/3}} k 1_{q^k \parallel n}$ instead of $\sum_{p\leq x^{1/3}} 1_{p \mid n}$. By this trick we can easily avoid thinking about numbers which are not square-free.
Are these two approaches equivalent in some easy way which I cannot see right now?
I would also like to thank you for your last answer.
[Oops, this case does need to be inserted, but it is easily dealt with (with $z = x^{1/8}$, it contributes $x^{7/8+o(1)}$ from trivial estimates). -T.]

Anonymous · 10 July 2019

Why is of $z := x^{\frac{1}{8}}$ in Chen's Theorem? Is it not the case that the closer z is to $z := x^{\frac{1}{3}}$ the better? I believe that using the same arguments and with same error term, one could set $z := x^{\frac{1}{k}}$, where k is the root of $\log \frac{k-2}{4} - \int_{\frac{1}{k}}^{\frac{1}{3}} \frac{\log(2-3x)}{x(1-x)} dx$, i.e, $k = 7.585\ldots$.

> **Terence Tao** · 10 July 2019
>
> In this post I am not attempting to optimise the exponents in order to keep the exposition as clean as possible. I believe the text of Friedlander-Iwaniec contains further improvements to the exponent for z, but I do not have the reference handy right now, and there may be further slight improvements in the literature.
> UPDATE: Section 25.6 of Friedlander-Iwaniec allows z to be as large as $x^{3/11}$. It may be that further improvements are still possible beyond this exponent.

Anonymous · 25 February 2020

Hi Prof Tao,
Just a quick question: why do we need eq. (31)? We seem to never use it in the proof. Apologies if I missed a clear usage of it.

> **Terence Tao** · 26 February 2020
>
> (31) is needed to establish some cases of (32), which is then used in the proof of Lemma 8.
> (One could cut out the middle man and just impose (32), but this formalism is closer to the general beta-sieve formalism one sees for instance in the text of Iwaniec-Kowalski or Friedlander-Iwaniec.)

Michelangelo Mecozzi · 23 March 2020

A tiny detail: when we define D_0 and z_0 should the values be swapped? With the current values we have $s_0 = \log D_0 / \log z_0 = 1/\varepsilon$ and I assume we want $s_0 = \varepsilon$ as in the proof.

> **Michelangelo Mecozzi** · 23 March 2020
>
> My bad, actually we want that and it wouldn't make sense to swap! I was working on the Exercise trying to show that the LHS of (34) is bounded by $f(s - 2\varepsilon^3) + o(1)$, this should be be $f(s - 2\varepsilon^2) + o(1)$ or I'm mistaken again? Doesn't make any difference in the calculations...
> *[Corrected, thanks – T.]*

Anonymous · 26 March 2020

Is it possible to get the following clarification: when you say (towards the end of the proof of the linear sieve) "Note that (29) also gives $p_r \geq (D/D_0^2)^{1/(r+2)}$" how can we deduce that?
[You are right, this is not the correct way to get a lower bound on p_r at this step. I will have to look at Friedlander-Iwaniec for the correct argument, but unfortunately due to the lockdown I will not be able to do so for a while. -T]

> **Terence Tao** · 5 April 2020
>
> I was finally able to retrieve my copy of Friedlander-Iwaniec. Theorem 2 is proven (in a stronger and more general form) as Theorem 11.12 from that paper. However the arguments are quite intricate, in particular the upper bounds on B_r are proven recursively. I now vaguely remember finding an alternate approach (based on using the Selberg counterexample weights) to write the notes here, but as you pointed out the lower bounding of p_r is incorrect. If one does compute what (29), (30) do give for lower bounds, they are more of the form $p_r \geq D^{1/C^r}$,

thus the exponent degrades exponentially in r rather than linearly. This still gives satisfactory control on the B_r when $r = o(\log \frac{1}{\varepsilon})$ but now there is an intermediate region $r \sim \log \frac{1}{\varepsilon}$ which is not currently covered by the above arguments. The issue is probably fixable, for instance by importing some of the recursive arguments from Friedlander-Iwaniec, but currently I will not have the time to address them and for now one should simply refer to Theorem 11.12 of Friedlander-Iwaniec for a proof of Theorem 2 in this post.



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[…] related to this, if one only sieves up to for some , the linear sieve (see e.g., Theorem 2 of this previous blog post) shows that one must have at least survivors, where can be given explicitly in the regime by the […]